Recap

Let's start by recalling the following algebraic properties,

Algebra properties

$$a+b=b+a, \quad (a+b)+c=a+(b+c), \quad a(b+c)=ab+ac,$$
$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}, \quad \frac{a}{b}\cdot\frac{c}{d}=\frac{ac}{bd}, \quad \frac{-a}{b}=-\frac{a}{b}=\frac{a}{-b}, \quad \frac{\frac{a}{b}}{\frac{c}{d}}=\frac{ad}{bc}$$
$$\sqrt{ab}=\sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}}=\frac{\sqrt{a}}{\sqrt{b}}, \quad \sqrt[n]{a}=a^{1/n}$$
$$a^2-b^2=(a-b)(a+b)$$

Laws of exponents

$$a^r\cdot a^s=a^{r+s}, \quad (a^r)^s=a^{rs}$$
$$\frac{a^r}{a^s}=a^{r-s}, \quad (ab)^r=a^r b^r$$
$$\left(\frac{a}{b}\right)^r=\frac{a^r}{b^r}, \quad b\neq 0$$

Laws of Logarithms

$$\log_a(AB)=\log_a(A)+\log_a(B)$$
$$\log_a\left(\frac{A}{B}\right)=\log_a(A)-\log_a(B)$$
$$\log_a\left(A^C\right)=C\log_a(A)$$

Inverse functions property

$$\left(f\circ f^{-1}\right)(x)=x=\left(f^{-1}\circ f\right)(x) \quad \to \quad \log_a(a^x)=x=a^{\log_a(x)}$$

Notation

Interval notation. To denote intervals we can use the following notation: $x\in(a,b)$, $x\in[a,b]$ or $a<x<b$. $a\leq x\leq b$. Remember that $(,)$ and $<$ are used for open boundaries, that is that the variable x can not take that value. Meanwhile, $[,]$ and $\leq$ are used for closed boundaries, that is that the variable x can take that value.

Class Activity

$$\text{Simplify}: \frac{x^2}{x-4}-\frac{x+1}{x+2}$$
$$\text{Factor}: x^3y-4xy$$
$$\text{Solve}: \frac{2x}{x+7}=\frac{2x-7}{x}$$
$$\text{Function compostion } (f\circ g)(x);$$
$$f(x)=x^2+3x-8;\ g(x)=3x-5$$

Interpretation of math

Let's assume that we can quantify the relation between money that is needed to print a book with a function $f(x)$. And we can quantify the relation between the money that we earn by selling a book $g(x)$. We can create an equation that allows us to find the money that is equal to selling and printing books.

Now, let's imagine that the following algebraic expression represents the situation above,

$$\frac{2x}{x+7} = \frac{2x-7}{x}.$$

Imagine that the term $\frac{2x}{x+7}$ quantifies the cost of printing a book and the term $\frac{2x-7}{x}$ quantifies the amount of money that is earned by selling books. We are going to find the value of x that satisfies this condition.

In order to find this condition, we can manipulate the expression using algebraic properties to the following,

$$7x - 49 = 0.$$

We have translated the problem of finding the intersection of two lines or functions into finding the roots of a polynomial. This "problem" conversion is very useful, because we know how to find roots of equations with algebraic methods. Which gives us that $x = 7$.

This condition, when we equate one function to another $f(x) = g(x)$, is normally referred to as the "*equilibrium condition*".

Intersections of a function.

Since a good strategy to find equilibrium conditions is to translate the equation into finding roots of polynomials, let's recall what a root of a polynomial is. A root of a polynomial is the value or set of values of the domain that makes the function equal to zero. That is, all the values of x that satisfy the condition $f(x) = 0$.

It is important to remember that a polynomial of degree n has n roots, assuming that the domain of the function, that is x, belongs to the set of complex numbers. However, if the domain is part of the set of real numbers, this is not true for all polynomials. For example, the function $f(x) = x^2 + 0.1$ does not have real roots. This indicates that not every equation with the expression $f(x) = g(x)$ can have a solution that belongs to the set of real numbers. And for an exam exercise, that is a valid and accurate response.

Class Activity

Solve the following equations

$$\frac{x^2+5x+4}{x^2-3x-4} = 0$$
$$x^3 - 3x^2 - 16x + 10 = 3x^2 + 11x + 10$$
$$3^{3x} + 3^x = 3^{2x+1}$$